

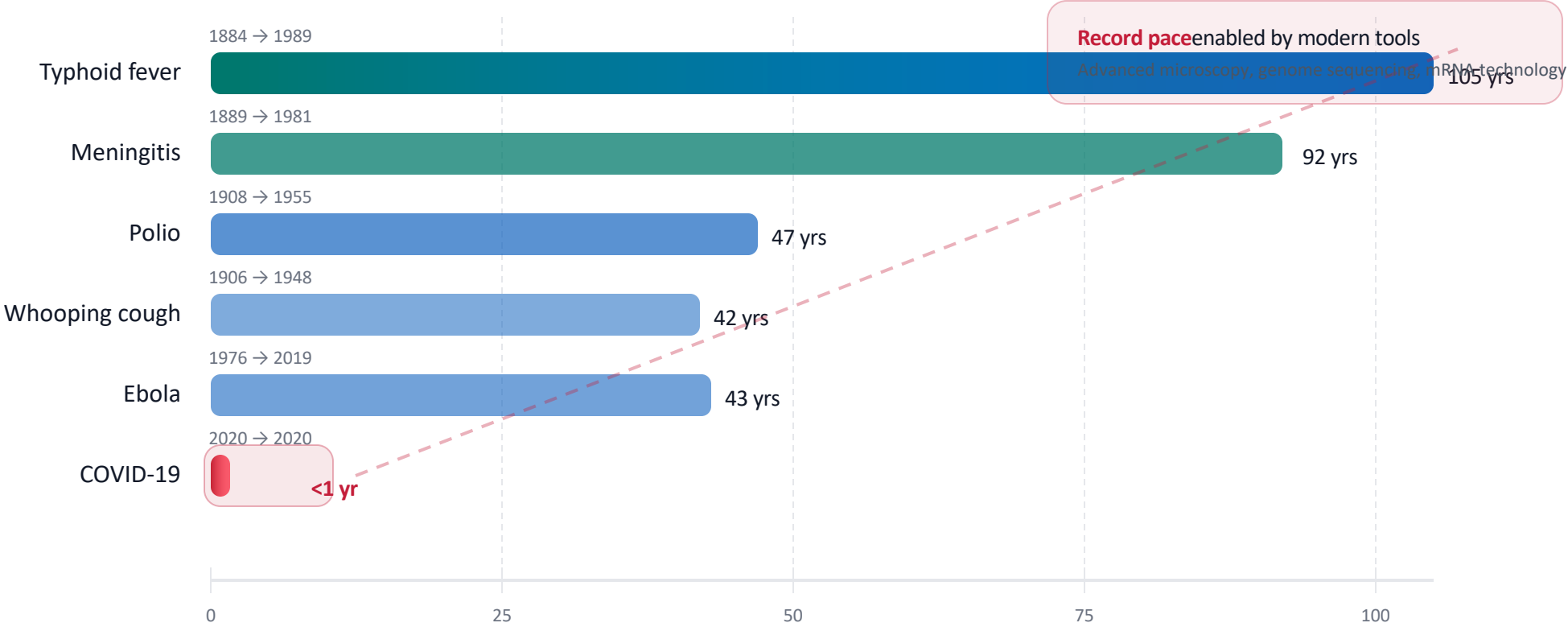
The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

Accelerating development (1796–2020) • Eliminating and reducing disease burden in the US

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

⚡ Pathogen-to-licensure timelines dropped from decades to <1 year for COVID-19.

Years from pathogen identification → vaccine licensure



100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the U

✔ Multiple diseases reached zero cases and zero deaths following vaccination.

Disease	Pre-vaccine cases	Post-vaccine cases	Case reduction	Pre-vaccine deaths	Death reduction
Diphtheria	158	0	100%	13.7	100% ✔
Smallpox	250	0	100%	2.9	100% ✔
Acute polio	141	0	100%	10	100% ✔
Measles	3,044	0.2	99.99%	2.5	100%
Pneumococcal	233	139	40.5%	24	31.3% ⚠
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%

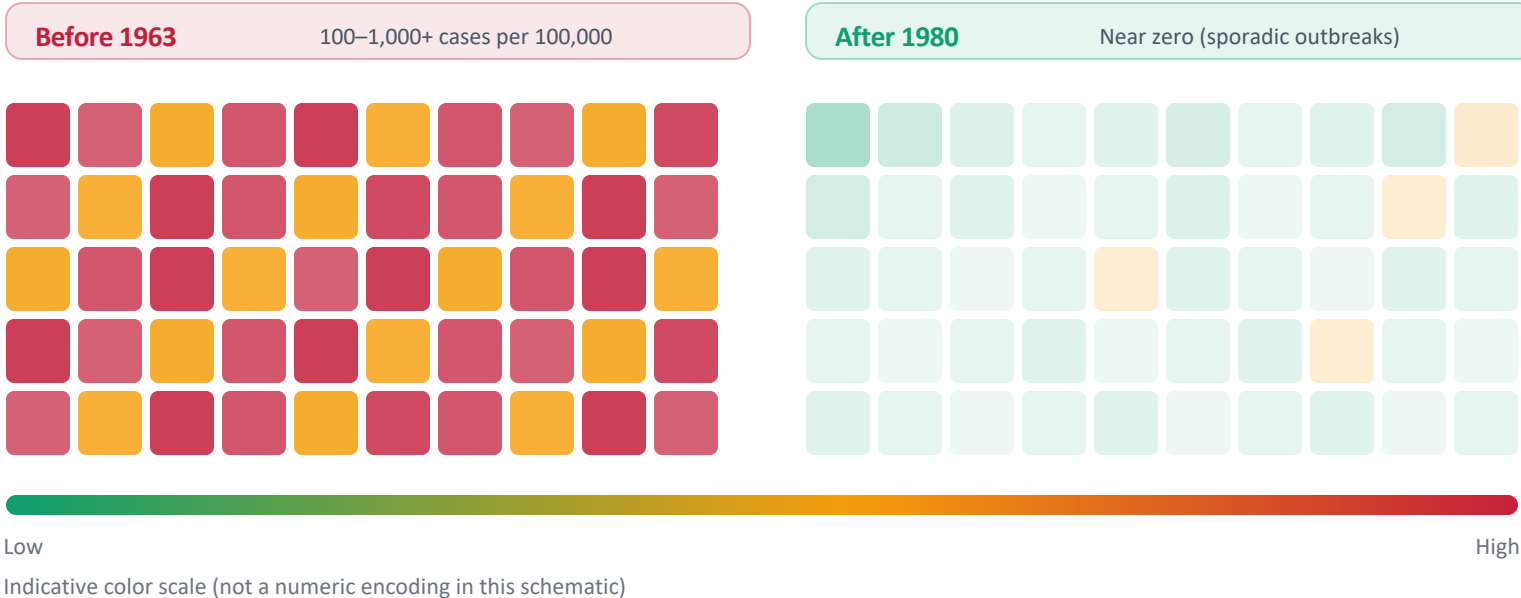
Values are per million per year, pre- vs. post-vaccine periods (Roush & Murphy, 2007; cited by Our World in Data).

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



Heatmap reference (1929–2022): high incidence before 1963 → near zero after the 1980 mandate.

Schematic state grid (50 states) — qualitative shift



Milestones

1963 First measles vaccine
John Enders

1971 MMR vaccine
Maurice Hilleman

1980 Kindergarten mandate
Vaccination required for entry

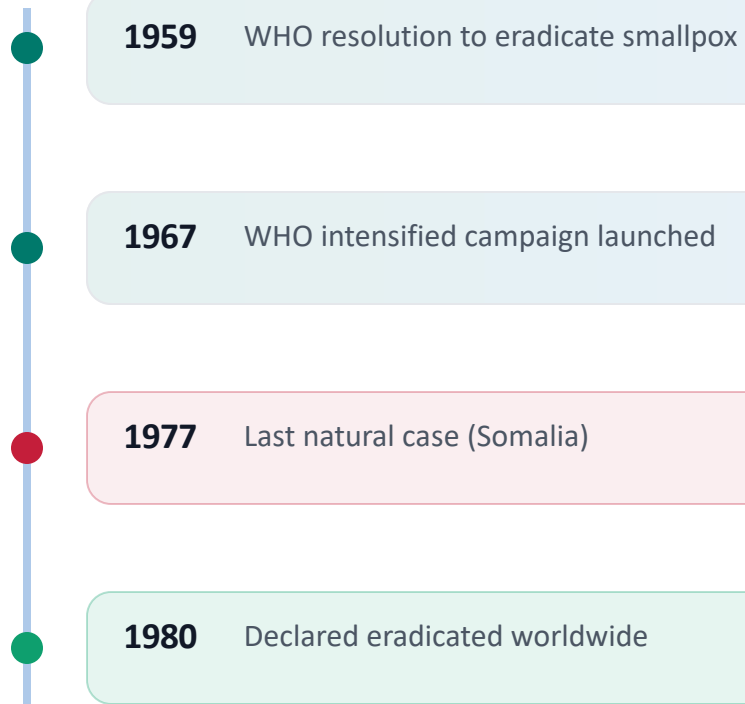
What the heatmap shows

- Before 1963: high incidence across nearly every state
- After 1980: near zero with sporadic outbreaks

Smallpox: The Only Human Disease Eradicated Through Vaccination

Global campaign milestones culminated in eradication certification in 1980.

Eradication timeline



Why eradication was possible

Infected only humans

No animal reservoir to sustain transmission

Symptoms distinguishable

Cases could be identified for targeted response

Vaccine cheap & effective

Scalable immunization campaigns worldwide

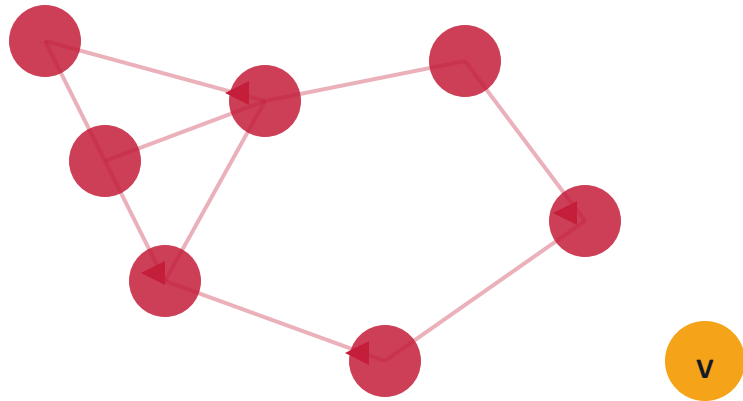
Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



When enough people are vaccinated, transmission chains break — shielding those who cannot be vaccinated.

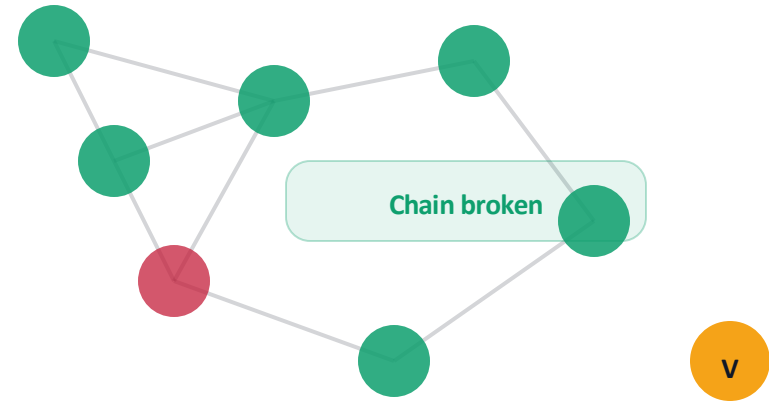
Low vaccination coverage

● Susceptible ● Vaccinated ● Vulnerable (cannot vaccinate)



Transmission spreads through connected susceptible people.

High vaccination coverage



Vaccinated people block transmission, indirectly protecting vulnerable individuals.



Protected groups include newborns, immunocompromised individuals, and cancer patients on chemotherapy.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

 Vaccine impact is uneven across diseases — and global access and trust still limit coverage.

Pneumococcal
40.5%
case reduction
31.3% death reduction

Hepatitis A
89%
case reduction

Acute hepatitis B
83.9%
case reduction

Barriers highlighted in the brief

Access gaps

- Not everyone has access to vaccines globally

Supply constraints

- Supply shortages put children at risk

Trust & understanding

- Poor scientific understanding can reduce trust and uptake

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Is Not Done

✓ Data-driven successes are clear — sustaining coverage and trust keeps progress durable.

- Vaccines achieved **100% elimination** of cases and deaths (US) for diphtheria, smallpox, and polio.
- Development timelines collapsed from **105 years** (typhoid) to **<1 year** (COVID-19).
- Measles dropped to near zero across all 50 states after the **1980** kindergarten mandate.
- Smallpox is the only human disease eradicated, certified in **1980** after a global campaign.
- Gaps persist for pneumococcal disease, hepatitis A, and hepatitis B — pointing to access and coverage needs.

Priority for health systems: Improve access, coverage, and trust in vaccination.